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Exploring how artificial intelligence capabilities impact corporate sustainability performance: Insights from Chinese manufacturing firms

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ABSTRACT

In the context of digital transformation, the driving mechanism of Artificial Intelligence capabilities (AIC) on corporate sustainability performance (CSP) and its organizational boundary requirements have emerged as a key topic in advancing digital sustainability, both in theory and practice. This study is based on dynamic capabilities theory (DCT) and sustainable development theory, and reveals the micro-mechanisms and boundary conditions of AIC driving CSP. By extending the traditional application of AI and dynamic capabilities from competitive advantage to sustainability, this study provides a novel theoretical lens for digital sustainability. This study empirically analyzes data from Chinese manufacturing enterprises through the PLS-SEM method. The findings reveal that AIC significantly enhances CSP through a tiered enabling mechanism, but prescriptive capabilities fail to achieve their intended effects. The type of manufacturing firm constitutes a core boundary condition, and there is significant variability in the sustainable performance of AIC exerted under traditional manufacturing and Industry 4.0 manufacturing firms. The impact of AIC on CSP is significantly weaker in traditional manufacturing firms than in Industry 4.0 manufacturing firms. By adopting a dynamic capabilities perspective, this study examines the role of DCT-based AIC in enhancing CSP, expanding the prevailing technical viewpoint on their interrelationship and providing a compelling justification for integrating AI into digital sustainability frameworks. These findings offer manufacturers actionable insights for harnessing AI technologies to achieve sustainable performance during digital transformation.

1. Introduction

Climate change is reshaping the global economic landscape at an unprecedented pace, pushing industrial enterprises-the primary sources of carbon emissions-to the forefront of sustainable transformation (Voosen, 2025). Global policy trends are rapidly shifting environmental performance from voluntary corporate social responsibility (CSR) to mandatory compliance requirements, such as China's "dual carbon" targets and the European Union's Corporate Sustainability Reporting Directive (CSRD) (Hernández et al., 2024; Bohnsack et al., 2022). However, traditional emission reduction methods such as energy substitution and process optimization have encountered significant technical bottlenecks. The rate of energy efficiency improvement lags far behind the level required to achieve carbon neutrality goals (Chu et al., 2024). This reality highlights the urgency of exploring innovative driving forces. Recently, digital technologies, especially artificial intelligence (AI), have played an essential role in the transition towards sustainable development (Schwaeke et al., 2025; Bickley et al., 2025; Chen et al., 2025). Therefore, improving industrial sustainability with AI has attracted increasing practical attention (Jiao et al., 2024).

Despite the increasing interest in utilizing AI to improve corporate sustainability performance (CSP), which is defined as a firm's integrated achievements in economic viability, environmental stewardship, and social responsibility (Elkington, 1997; Schaltegger and Wagner, 2011), current studies on the relationship between AI and CSP primarily emphasize qualitative research and lack empirical evidence (Zhou et al., 2025; Schwaeke et al., 2025). Among them, some studies taking the technology perspective elucidate that AI can acquire new knowledge to support managers' rational decision-making, improve resource utilization efficiency (George et al., 2020), and reduce environmental pollution, thus enhancing CSP (Wang et al., 2024; Hussain et al., 2024). However, scholars taking the capability perspective hold that artificial

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